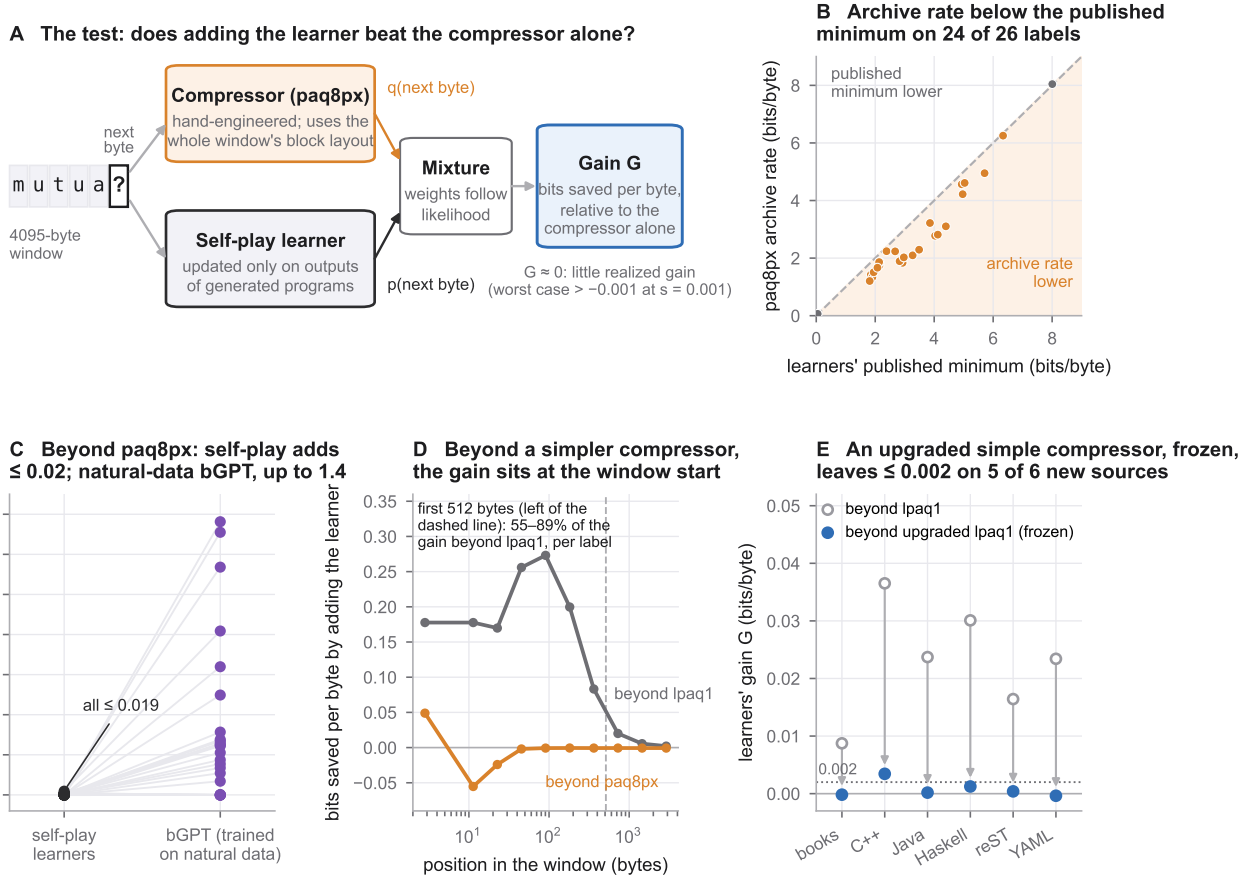

A The test: does adding the learner beat the compressor alone?
Compressor (paq8px)
hand-engineered; uses the whole window's block layout
q(next byte)
next byte
m u t u a ?
4095-byte window
Self-play learner
updated only on outputs of generated programs
Mixture
weights follow likelihood
Gain G
bits saved per byte, relative to the compressor alone
p(next byte)
G ≈ 0: little realized gain
(worst case > −0.001 at s = 0.001)
B Archive rate below the published minimum on 24 of 26 labels
published minimum lower
archive rate lower
paq8px archive rate (bits/byte)
learners' published minimum (bits/byte)
C Beyond paq8px: self-play adds ≤ 0.02; natural-data bGPT, up to 1.4
gain beyond paq8px, G (bits/byte)
all ≤ 0.019
self-play learners
bGPT (trained on natural data)
D Beyond a simpler compressor, the gain sits at the window start
bits saved per byte by adding the learner
first 512 bytes (left of the dashed line): 55–89% of the gain beyond lpaq1, per label
beyond lpaq1
beyond paq8px
position in the window (bytes)
E An upgraded simple compressor, frozen, leaves ≤ 0.002 on 5 of 6 new sources
learners' gain G (bits/byte)
beyond lpaq1
beyond upgraded lpaq1 (frozen)
0.002
books
C++
Java
Haskell
reST
YAML